

Assignment / Tutorial / ~~Objective Type Question Paper~~

KPR Institute of Engineering and Technology (Autonomous) Avinashi Road, Arasur, Coimbatore - 641 407			Dept.:	CSE
Course Code & Title	:	U21CS501 Web Technology	Ac.Yr :	2026-2027 ODD
Year	:	III	Issue Date	10.08.2026
Semester / Section	:	V / C	Submission Date	17.08.2026
Assignment	:	I	Maximum Marks	40

Task Instructions:

1. Individual Project
2. Design and develop a web application for any real-time industrial use case.
3. Each student must select a unique application/use case.
4. The application must demonstrate the concepts specified under CO1, CO2 and CO3.
5. Students may use Bolt AI for development assistance, but every student must clearly understand and explain the complete code during evaluation/viva.

Q.No	Answer All Questions	M	BT	CO
1	1. Explain the URL and its structure, including how the browser parses the URL. 2. Explain the DNS resolution process from domain name to IP address and how the browser communicates with the web server. 3. Using Chrome Developer Tools → Network, capture and explain the HTTP Request and HTTP Response generated by your application. 4. Include suitable screenshots showing the Request URL, Method, Status Code, Headers and Response of your webpage. 5. Analyze the web performance of your application using Chrome Developer Tools/Lighthouse and include screenshots of important performance results. Sample Applications: (Not limited to) 1. AI-Enabled Smart Manufacturing & Predictive Maintenance System Users: Admin Production Manager Maintenance Engineer 2. Intelligent Healthcare Appointment & Patient Care Management System Users: Hospital Admin Doctor Patient 3. Smart Logistics, Fleet & Real-Time Shipment Tracking System Users: Logistics Admin Fleet Manager Delivery Executive 4. Enterprise Warehouse Inventory & Supply Chain Management System Users: Warehouse Admin Inventory Manager Warehouse Operator	16	AP	CO1

	5. Industrial Equipment Service & Maintenance Management System Users: System Admin Maintenance Manager Service Technician ALTERNATE TASK: 1. Infosys Springboard – HTML Course Completion Certification			
2	Task: 1. Create a project directory with all essential HTML, CSS and JavaScript files. 2. Use essential HTML5 elements such as structure/semantic tags, headings, paragraphs, text formatting, links, images, lists, tables, forms and suitable multimedia/embedding elements. 3. Apply external CSS using selectors, box model, layouts, backgrounds, borders and responsive design. 4. Include a Login Page with exactly three hard-coded user roles and redirect each role to a different dashboard with role-specific modules/functions. 5. Design all pages based on the selected industrial workflow, making the application realistic and relevant to the chosen industry. ALTERNATE TASK: Infosys Springboard – CSS Course Completion Certification	16	AP	CO2
3	Task: 1. Implement JavaScript form validation for name/text, phone number, email and other relevant fields using Regular Expressions (RegEx). 2. Create and use JavaScript functions and events to handle user interactions. 3. Perform DOM operations to dynamically change or update webpage content. 4. Display meaningful validation/error messages and update form or application data based on user actions. 5. Implement relevant interactive features such as searching/filtering records, showing/hiding content, updating status or modifying dashboard information. ALTERNATE TASK: Infosys Springboard – Java Script Course Completion Certification (During Second Assessment 8 More marks can be added)	8	AP	CO3

Submission Requirements

1. Prepare the .pdf file of the complete developed web application, including all HTML, CSS, JS, images and other required files.
2. Host the application on GitHub by creating a GitHub repository and publishing the website.
3. Fill in the Google Form and submit the required details, including the GitHub Repository URL, Hosted Website URL, and .pdf file.

Evaluation Rubric (20 Marks)

Criteria	Marks
Application Uniqueness	10
Case study analysis on HTTP Request and Response along with URL and DNS with appropriate screen shorts from Chrome Dev tools	10
Design and application quality of Static Pages HTML and CSS	10
Quality of Client-Side validation including DOM, Form validation, Event handling	10
Total	40



Name and Signature of Course In-charge



Approved by Head of the Department